

ASSESSMENT TOOL 2
ITA0609 - Machine Learning

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1. Compare Case-Based Learning and Locally Weighted Regression Using a Practical Example Such as Recommendation Systems

Introduction

Case-Based Learning (CBL) and Locally Weighted Regression (LWR) are important machine learning approaches that learn from previously observed examples rather than constructing a single fixed global model. Both methods are useful when the relationship between input and output changes across different situations. However, they differ in their learning objectives and prediction mechanisms. Case-Based Learning retrieves and reuses solutions from similar previous cases, whereas Locally Weighted Regression constructs a local regression model around the current query point. A recommendation system is a suitable practical application for comparing these two approaches because recommendations can be generated by analyzing similarities between users and their previous preferences.

Practical Example: Movie Recommendation System

Consider an online movie recommendation system that recommends movies to users based on their interests and previous viewing behavior. The system stores information such as age, preferred genres, average rating, number of movies watched, and ratings given to previous movies. When a new user enters the system, the recommendation engine compares the user's characteristics with historical user data and predicts suitable movies.

User	Action Preference	Comedy Preference	Drama Preference	Average Rating	Recommended Movie
U1	0.90	0.50	0.20	4.5	Movie A
U2	0.80	0.60	0.30	4.2	Movie A
U3	0.20	0.80	0.70	3.8	Movie B
U4	0.90	0.70	0.20	4.4	Movie A

Suppose a new user has an action preference of 0.85, comedy preference of 0.60, drama preference of 0.25, and an average rating of 4.3. The system can compare this user with previous users and determine the most appropriate recommendation.

Case-Based Learning

Case-Based Learning solves a new problem by finding previous cases that are similar to the current problem and reusing the solutions associated with those cases. In a recommendation system, a case consists of a user's characteristics, previous interactions, selected products or movies, and the outcome of the recommendation. The system searches its case library whenever a new recommendation is required.

The standard Case-Based Learning process consists of four stages: **Retrieve, Reuse, Revise, and Retain**. During the Retrieve stage, the system identifies historical cases that are most similar to the new user. During Reuse, the solution associated with the retrieved cases is used to generate a recommendation. During Revise, the recommendation can be modified based on additional information or user feedback. During Retain, the new user interaction is stored as a new case so that it can be used for future recommendations.

Similarity Measurement

Similarity between users can be calculated using Euclidean distance for numerical features. The Euclidean distance between two users (X) and (Y) is given by:

$$[d(X,Y)=\sqrt{\sum_{i=1}^n(X_i-Y_i)^2}]$$

A smaller distance indicates that the users have more similar characteristics. For example, if the new user has preferences similar to U1, U2, and U4, these users become the most relevant cases for generating the recommendation. Since all three users were recommended Movie A, the system would recommend Movie A to the new user.

Locally Weighted Regression

Locally Weighted Regression is a local learning method used mainly for predicting continuous values. Instead of giving equal importance to every training observation, LWR assigns higher weights to observations that are closer to the current query point. Therefore, users with highly similar characteristics have greater influence on the predicted rating.

A commonly used Gaussian weighting function is:

$$[w_i=\exp\left(-\frac{d(x,x_i)^2}{2\tau^2}\right)]$$

Here, $(d(x,x_i))$ represents the distance between the new user and an existing user, while (τ) is the bandwidth parameter that controls the locality of the model.

Example of Locally Weighted Prediction

Suppose three similar users have given the following ratings to a particular movie:

User	Weight	Rating
U1	0.90	4.8
U2	0.75	4.2
U3	0.40	3.5

The predicted rating can be calculated using a weighted average:

$$\hat{y} = \frac{\sum w_i y_i}{\sum w_i}$$

$$\hat{y} = \frac{(0.90)(4.8) + (0.75)(4.2) + (0.40)(3.5)}{0.90 + 0.75 + 0.40}$$

$$\hat{y} = \frac{4.32 + 3.15 + 1.40}{2.05}$$

$$\hat{y} = 4.33$$

Therefore, the predicted rating is approximately **4.33 out of 5**, indicating that the movie is likely to be recommended.

Handling New Data

Case-Based Learning handles new data by storing new user experiences as additional cases. For example, if a customer initially prefers action movies but later begins watching more science-fiction movies, this new behavior can be stored as a new case. Future recommendations can then use the updated information.

Locally Weighted Regression also handles new data effectively because predictions are generated using currently available observations. New observations can be incorporated into the dataset without completely rebuilding a global regression model. However, the system must efficiently search the dataset to identify observations close to the new query.

Adaptability to Changing Behaviour

Case-Based Learning has strong adaptability because every new experience can be retained in the case library. Recent user behavior can therefore influence future recommendations. This is particularly useful for recommendation systems because user interests can change over time.

LWR is also adaptive because it focuses on observations that are locally similar to the current input. If the relationship between user characteristics and ratings changes, the local model can adjust according to the latest relevant observations. However, the quality of adaptation depends on the availability of sufficient recent and relevant data.

Computational Efficiency

Case-Based Learning has very low training cost because it mainly stores previous cases rather than learning a complex mathematical model. However, prediction can become computationally expensive when the number of stored cases becomes very large because the system must search for similar cases.

Locally Weighted Regression also has relatively low training cost, but its prediction process can be computationally expensive. For each new query, distances must be calculated, weights must be assigned, and a local regression model must be constructed. Therefore, both approaches shift a significant amount of computational work from training to prediction.

Comparison

Criterion	Case-Based Learning	Locally Weighted Regression
Learning Type	Instance-based learning	Local regression
Main Operation	Retrieve similar cases	Construct local model
Typical Output	Existing solution/recommendation	Continuous prediction
Training Cost	Very Low	Very Low
Prediction Cost	Can be High	Can be High
Adaptability	Very High	High
New Data	Stored as new cases	Incorporated into local prediction
Interpretability	Very High	High
Suitable Application	Recommendation and decision reuse	Numerical prediction
Main Parameter	Similarity measure	Bandwidth

Advantages of Case-Based Learning

Case-Based Learning is simple, highly interpretable, and easy to update. The system can explain its recommendation by identifying similar users and showing what worked for them previously. It is particularly useful when previous solutions can be directly reused for new situations.

Advantages of Locally Weighted Regression

Locally Weighted Regression is useful when the output is continuous and the relationship between input and output changes across different regions of the dataset. It can capture local nonlinear behavior without requiring a complex global model.

Conclusion

Case-Based Learning and Locally Weighted Regression are both adaptive learning approaches, but they are suited to different types of recommendation problems. Case-Based Learning is more appropriate when recommendations can be directly derived from similar historical users and when explainability is important. Locally Weighted Regression is more suitable when the system needs to predict a continuous value such as a user's expected rating. Both approaches can incorporate new data without expensive global retraining, but their prediction cost increases with dataset size. For a practical recommendation system where previous customer experiences are highly valuable, **Case-Based Learning is the more suitable approach**, while LWR is useful when numerical ratings or preference scores must be predicted.

2. Compare Locally Weighted Regression and Radial Basis Function Networks for Function Approximation Tasks

Introduction

Function approximation is the process of learning an unknown mathematical relationship between input variables and an output from a collection of observed examples. **Locally Weighted Regression (LWR)** and **Radial Basis Function (RBF) Networks** are two machine learning approaches capable of approximating nonlinear functions. Both methods use the concept of locality, meaning that observations or learned centers close to an input have a stronger influence on the prediction. However, LWR performs local regression separately for each query, whereas an RBF Network learns a set of radial basis functions during training and uses them to approximate the entire function.

Example Dataset

Consider a nonlinear dataset generated approximately from the function:

$$\begin{aligned} &[\\ &y = \sin(x) + \epsilon \\ &] \end{aligned}$$

where (ϵ) represents a small amount of random noise.

x	Observed y
0.0	0.05
0.5	0.47
1.0	0.86
1.5	1.02
2.0	0.89
2.5	0.63
3.0	0.16
3.5	-0.37
4.0	-0.72
4.5	-0.98
5.0	-0.91
5.5	-0.68
6.0	-0.25

The objective is to learn the nonlinear relationship between (x) and (y).

Locally Weighted Regression

Locally Weighted Regression is a non-parametric learning technique that constructs a local model around each query point. When a new input is provided, LWR determines the distance between that input and each training observation. Nearby observations receive high weights, while distant observations receive low weights.

A Gaussian weighting function can be represented as:

$$w_i = \exp\left(-\frac{(x-x_i)^2}{2\tau^2}\right)$$

where (x) is the query point, (x_i) is a training observation, and (τ) is the bandwidth parameter.

For example, when predicting (y) at (x=2.0), observations around (x=1.5), (x=2.0), and (x=2.5) receive high weights because they are close to the query. Observations at (x=5.0) or (x=6.0) receive very low weights because they are far away.

Radial Basis Function Network

An RBF Network is a type of neural network consisting of an input layer, a hidden layer containing radial basis functions, and an output layer. Each hidden neuron has a center and measures the distance between the input and that center. A Gaussian RBF is commonly represented as:

$$\phi_j(x) = \exp\left(-\frac{(x-c_j)^2}{2\sigma_j^2}\right)$$

where (c_j) is the center of the (j)-th radial basis function and (σ_j) represents its width.

The final output of an RBF Network is:

$$\hat{y} = \sum_{j=1}^m w_j \phi_j(x) + b$$

For the given dataset, centers could be selected at approximately (0), (1.5), (3), (4.5), and (6). Each RBF neuron responds strongly when the input is close to its center. The weighted combination of these responses produces the final approximation of the nonlinear function.

Difference in Locality

LWR is highly local because it performs a new local regression for every query point. Its prediction depends directly on the nearby training observations. If the query changes, the set of important observations and their weights also change.

RBF Networks also represent locality, but the locality is captured through learned centers and basis functions. After training, the same set of centers and weights is used for future predictions. Therefore, RBF Networks provide a more structured and reusable representation of the input space.

Computational Cost

LWR has very little training cost because it does not construct a permanent global model. However, it has a relatively high prediction cost because distances and weights must be calculated for the training observations whenever a new prediction is requested. This becomes expensive when the dataset contains a very large number of observations.

RBF Networks require greater computational effort during training because the centers, widths, and output weights must be learned. However, once the network has been trained, prediction is faster because only the learned RBF neurons need to be evaluated.

Operation	LWR	RBF Network
Training Cost	Low	Moderate to High
Prediction Cost	High	Low
Model Storage	Training Data	Centers and Weights
Updating	Easy	Requires retraining/update
Large Dataset	Less efficient	More efficient after training

Smoothness of Predictions

The smoothness of LWR predictions depends strongly on the bandwidth parameter (τ). When the bandwidth is very small, the model concentrates heavily on nearby observations. This can cause the model to follow noise and produce irregular predictions. When the bandwidth is increased, more observations contribute to the prediction, resulting in a smoother curve but potentially losing important local variations.

RBF Networks generally produce smooth predictions because multiple Gaussian basis functions overlap across the input space. The width of each RBF controls the region over which a center influences the prediction. Appropriate selection of the number of centers and their widths allows the network to approximate complex nonlinear functions smoothly.

Handling Noise

LWR can reduce the influence of distant noisy observations because they receive smaller weights. However, a noisy observation located close to the query point can still receive a large weight and influence the prediction significantly.

RBF Networks can provide smoother approximations and reduce the effect of random noise when appropriate regularization, basis-function widths, and network complexity are selected.

An excessive number of RBF neurons can cause overfitting, while too few neurons may result in underfitting.

Effect of Dataset Size

For a small dataset, LWR can be an effective choice because it requires little training and can immediately use the available observations. As the dataset becomes larger, the repeated distance calculations required by LWR become increasingly expensive.

RBF Networks are more suitable when many predictions need to be generated after an initial training phase. The network compresses the information contained in the original dataset into learned centers and weights, making inference more efficient.

Comparison of LWR and RBF Networks

Criterion	Locally Weighted Regression	RBF Network
Learning Approach	Non-parametric local learning	Neural network
Locality	Query-dependent	Center-dependent
Training	Minimal	Required
Prediction	Computationally expensive	Faster
Smoothness	Controlled by bandwidth	Controlled by RBF widths
Noise Handling	Distance-based weighting	Smooth basis functions
Adaptability	Very High	Moderate
Scalability	Lower for large datasets	Better after training
Interpretability	Relatively High	Moderate
Best Use	Small/adaptive datasets	Repeated nonlinear prediction

Advantages of Locally Weighted Regression

LWR is simple, flexible, and highly adaptive. It does not require the entire dataset to follow one global mathematical relationship. It is capable of capturing local nonlinear behavior and can immediately incorporate new observations. It is particularly suitable when the dataset is relatively small and the underlying function changes significantly across different regions.

Advantages of RBF Networks

RBF Networks are powerful nonlinear function approximators that can learn complex relationships between inputs and outputs. They provide smooth predictions, have relatively fast inference after training, and can represent nonlinear functions using a limited number of learned centers. They are particularly useful when many predictions must be generated from a relatively stable dataset.

Conclusion

Locally Weighted Regression and Radial Basis Function Networks are both effective methods for nonlinear function approximation, but they differ in how locality and computation are handled. LWR performs a query-specific local regression and therefore has very low training cost and high adaptability, but prediction becomes expensive as the dataset grows. RBF Networks require more computational effort during training but provide faster prediction after learning a set of radial basis functions. LWR is more appropriate for small datasets and applications requiring frequent adaptation, whereas RBF Networks are better suited for larger, relatively stable datasets requiring repeated and smooth predictions. Therefore, the choice between the two methods depends primarily on **dataset size, required adaptability, computational resources, and the desired smoothness of the approximation.**